

Supplementary Appendix

Supplement to: Cho BC, Lu S, Felip E, et al. Amivantamab plus lazertinib in previously untreated *EGFR*-mutated advanced NSCLC. N Engl J Med. DOI: 10.1056/NEJMoa2403614

This appendix has been provided by the authors to give readers additional information about the work.

Amivantamab Plus Lazertinib in Treatment-naïve *EGFR*-mutated Advanced NSCLC

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List of MARIPOSA Investigators

Principal Investigator	Clinical Site
Andres Aguilar	Hospital Universitari Quiron Dexeus
Azura Rozila Ahmad	Beacon Hospital Sdn. Bhd.
Jorge Alatorre-Alexander	Health Pharma Professional Research
Sabina Aleixo	Hospital Evangélico de Cachoeiro de Itapemirim
Adlinda Alip	University of Malaya Medical Centre
Rosa Alvarez	Hospital General Universitario Gregorio Marañón
Sara Alves	Instituto Português de Oncologia do Porto
Timur Andabekov	Oncology Medical Clinics AV Medical group
Antonio Araujo	Centro Hospitalar Universitario do Porto, EPE
Cagatay Arslan	Medical Point
Mehmet Artac	Necmettin Erbakan University Meram Medical Faculty
Koichi Azuma	Kurume University Hospital
Clarissa Baldotto	Instituto D'Or de Pesquisa e Ensino (IDOR)
Joaquim Bosch Barrera	Institut Català D'Oncologia Girona
Cristina Bayona	Hospital Universitario de Burgos
Benjamin Besse	Institut Gustave Roussy
Bonne Biesma	Jeroen Bosch Ziekenhuis
Ahmet Bilici	Medipol Mega University Hospital
Ana Blasco	Hospital General Universitario de Valencia
Igor Bondarenko	Dnipropetrovsk State Medical Academy, Dnipropetrovsk City Multifield Clinical Hospital # 4
Lisenka Boom	Ziekenhuis St Jansdal
Samantha Bowyer	Sir Charles Gairdner Hospital

Emilio Bria	IRCCS Gemelli
Carlos Alberto Brocca	Hospital Privado Centro Medico de Cordoba
Raffaele Califano	Christie NHS Foundation Trust
Bangwei Cao	Beijing Friendship Hospital, Capital Medical University
Reyes Bernabe Caro	Hospital Virgen del Rocio
Gee-Chen Chang	Chung Shan Medical University Hospital
Chaiyut Charoentum	Chiang Mai University
Jianhua Chen	Hunan Cancer Hospital
Yuan Chen	Tongji Hospital, Tongji Medical College, Huazhong University of Science & Technology
Ying Cheng	Jilin Cancer Hospital
Byoung Chul Cho	Severance Hospital, Yonsei University Health System
Yamil Lopez Chuken	I Can Oncology Center
Timucin Cil	Adana City Hospital
Niels Claessens	Rijnstate Ziekenhuis
Benoit Colinet	Grand Hopital de Charleroi, Notre Dame
Diego Cortinovis	San Gerardo Hospital
Alexis Cortot	Institute Coeur Poumon
Marcio Valerio Costa	UPCO Unidade de Pesquisa Clinica em Oncologia
Enric Carcereny Costa	Institut Català D'Oncologia-Badalona
Sophie Cousin	Institut Bergonié
Felipe José Silva Melo Cruz	Núcleo de Pesquisa São Camilo
Marcia Cruz-Correa	Pan American Center for Oncology Trials LLC
Pongwut Danchaivijitr	Siriraj Hospital
Manolo D'Arcangelo	Ospedale S. Maria Delle Croci

Javier De Castro	Hospital Universitario La Paz
Gilberto De Castro Junior	Fundacao Faculdade de Medicina, Instituto do Cancer do Estado de Sao Paulo
Angelo Delmonte	Istituto Scientifico Romagnolo per lo Studio e la Cura dei Tumori
Filippo De Marinis	Istituto Europeo di Oncologia
Ingel Demedts	Algemeen Ziekenhuis Delta
Christophe Doms	UZ Leuven
Rafal Dziadziuszko	Uniwersyteckie Centrum Kliniczne
Ekkehard Eigendorff	Zentralklinik Bad Berka GmbH
Maria Isabel Enriquez	CIMOVA, Morals Vargas Centro de Investigación SC
Bulent Erdogan	Trakya University Medical Faculty
Jian Fang	Beijing Cancer Hospital
Yong Fang	Sir Run Run Shaw Hospital, Zhejiang University School of Medicine
Enriqueta Felip	Hospital Universitario Vall d'Hebron
Miguel Fernandez de Sanmamed	Clínica Universidad de Navarra
Carlos Gil Moreira Ferreira	Oncoclínicas Rio de Janeiro S.A.
Juergen Fischer	Onkologische Schwerpunktpraxis
Martin Forster	UCL Cancer Institute
Fabio Franke	Oncosite, Centro de Pesquisa Clínica em Oncologia Ltda
Helano Freitas	Fundacao Antonio Prudente, A.C. Camargo Cancer Center
Daichi Fujimoto	Wakayama Medical University Hospital
Shirish Gadgil	Henry Ford Hospital

Yousuf Gaffar	Maryland Oncology Hematology, PA
Gabriella Galffy	Tudogyogyintezet Torokbalint
Sandip Ganguly	Tata Medical Center
Beili Gao	Ruijin Hospital, Shanghai Jiao Tong University
Maria del Rosario Garcia Campelo	Hospital Universitario A Coruña
Lizbett Vanessa García Montes	Mexico Centre for Clinical Research, S.A. de C.V.
Ana Caroline Gelatti	Uniao Brasileira de Educação e Assistencia, Hospital Sao Lucas da PUCRS
Christian Gessner	POIS Leipzig GbR
Nicolas Girard	Institut Curie
Maya Gottfried	Meir Medical Center
Frank Griesinger	Pius-Hospital Oldenburg
Mahmut Gumus	T.C. Saglik Bakanligi Goztepe Prof. Dr. Suleyman Yalcin Sehir Hastanesi
Zhongliang Guo	Shanghai East Hospital
Vanesa Gutiérrez Calderón	Hospital Regional Universitario de Malaga
Zhigang Han	Cancer Hospital of Xinjiang Medical University
Direndra Hasmucrai	Centro Hospitalar Lisboa Norte EP, Hospital Pulido Valente
Osamu Hataji	Matsusaka Municipal Hospital
Hidetoshi Hayashi	Kindai University Hospital
Carole Helissey	HIA Begin
Osvaldo Hernandez	Oncologia Integral Satellite
Hwoei Fen Soo Hoo	Hospital Pulau Pinang

Soon-Hin How	Hospital Tengku Ampuan Afzan
Te-Chun Hsia	China Medical University Hospital
Yi Hu	Chinese PLA General Hospital
Jen-Yu Hung	Kaohsiung Medical University, Chung-Ho Memorial Hospital
Eiki Ichihara	Okayama University Hospital
Dolores Isla	Hospital Clínico Universitario Lozano Blesa
Minish Jain	Noble Hospital Pvt Ltd
Mei Ji	The First People's Hospital of Changzhou
Robert Jotte	Rocky Mountain Cancer Centers
Rosalyn Juergens	Juravinski Cancer Centre
Diego Lucas Kaen	Centro Oncologico Riojano Integral
EunJoo Kang	Korea University Guro Hospital
Jin Hyoung Kang	The Catholic University of Korea, Seoul St. Mary's Hospital
Tetsuji Kawamura	National Hospital Organization Himeji Medical Center
Andrzej Kazarnowicz	Warminsko-Mazurskie Centrum Chorob Pluc w Olsztynie
Sarah Khan	Nottingham University Hospitals NHS Trust
Sang-We Kim	Asan Medical Center
Yu Jung Kim	Seoul National University Bundang Hospital
Martin Kimmich	Robert-Bosch-Krankenhaus - Klinik Schillerhoehe
Zsolt Kiraly	Veszprém Megyei Tudógyógyintézet
Nikolay Kislov	Yaroslavl Regional Clinical Oncology Hospital
Satoru Kitazono	The Cancer Institute Hospital of JFCR
Ernesto Korbenfeld	Centro Oncológico Korben
Dariusz Kowalski	Narodowy Instytut Onkologii im. Marii Skłodowskiej-Curie, Panstwowy Instytut Badawczy

Ruben Kowalyszyn	Clínica Viedma
Shoichi Kuyama	National Hospital Organization Iwakuni Clinical Center
Jong-Seok Lee	Seoul National University Bundang Hospital
Kang-Yun Lee	Taipei Medical University Shuang Ho Hospital
Ki-Hyeong Lee	Chungbuk National University Hospital
Se-Hoon Lee	Samsung Medical Center
SungSook Lee	Inje University Haeundae Paik Hospital
Yun-Gyoo Lee	Kangbuk Samsung Hospital
Oleg Leonov	Omsk Clinical Oncology Dispensary
Viktoriya Leonova	Municipal Nonprofit Enterprise “Regional Center of Oncology”
Jianguo Li	Sun Yat-Sen Memorial Hospital Sun Yat-sen University
Juan Li	Sichuan Cancer Hospital
Wen Li	Second Affiliated Hospital, School of Medicine, Zhejiang University
Yuping Li	The First Affiliated Hospital of Wenzhou Medical University
Chun Sen Lim	Hospital Sultan Ismail
Chien-Chung Lin	National Cheng Kung University Hospital
Baogang Liu	Harbin Medical University Cancer Hospital
Caigang Liu	Shengjing Hospital of China Medical University
Shun Lu	Shanghai Chest Hospital
Zhe Liu	Beijing Chest Hospital, Capital Medical University
Dongqing Lv	Taizhou Hospital of Zhejiang Province
Melanie MacKean	Edinburgh Cancer Centre Western General

Tadashi Maeda	National Hospital Organization Yamaguchi Ube Medical Center
Margarita Majem	Hospital de la Santa Creu i Sant Pau
Claudio Martin	Instituto Alexander Fleming
Marina Matrosova	Nizhny Novgorod Regional Oncological Dispensary
Shingo Matsumoto	National Cancer Center Hospital East
Ofer Merimsky	Tel-Aviv Sourasky Medical Center
Tomoyasu Mimori	Juntendo University Hospital
Daniel Misch	Helios Klinikum Emil von Behring GmbH
Krishna Mohan M. V. T	Basavatarakam Indo-American Hospital
Masahide Mori	National Hospital Organization Osaka Toneyama Medical Center
Denis Moro-Sibilot	CHU de Grenoble, Hopital Albert Michallon
Josiane Mourao Dias	Fundacao Pio XII
Sanjay Mukhedkar	St John of God Hospital Murdoch
Govardhanan Nagaiah	Arizona Oncology Associates, PC – HAL
Rajnish Nagarkar	HCG Manavta Cancer Centre
Thomas Newsom-Davis	Chelsea And Westminster Hospital
Danny Nguyen	City of Hope Long Beach Elm
Naoyuki Nogami	Ehime University Hospital
Silvia Novello	Aou San Luigi Gonzaga
Ken O'Byrne	Princess Alexandra Hospital
Svetlana Odintsova	Current Medical Technologies
Francesca Ogliari	IRCCS Ospedale San Raffaele
Mikhail Osipov	Leningrad Regional Oncology Dispensary

Yurii Ostapenko	National Cancer Institute
Gyula Ostoros	Orszagos Koranyi Tbc es Pulmonologiai Intezet
Sai-Hong Ignatius Ou	University of California Irvine
Scott Owen	McGill University Health Centre
Yoshitaka Oyamada	National Hospital Organization Tokyo Medical Center
Mustafa Ozguroglu	Istanbul Universitesi Cerrahpasa tip Fakultesi ic Hastaliklari Anabilim Dali Medikal Onkoloji bd
Ozgur Ozyilkan	Başkent University Medical Faculty, Adana Application and Research Center
Charles Andréé Joseph Pádua	Cetus Oncologia
Zsolt Papai-Szekely	Fejer Varmegyei Szent Gyorgy Egyetemi Oktatokorhaz
Sagun Parakh	Austin Hospital
Barbara Parente	Hospital CUF Porto
Hector Jose Soto Parra	Azienda Ospedaliero Univ. Policlinico Gaspare Rodolico
Antonio Passaro	Istituto Europeo di Oncologia
Luis Paz-Ares	Hospital Universitario 12 de Octubre
Julio Peguero	Oncology Consultants, Texas
Kirstin Perdrizet	William Osler Health System
Marina Peregudova	MCK
Mauricio Peressoni	Ynova Pesquisa Clinica
Idit Peretz	Rabin Medical Center
Eric Pichon	CHU Bretonneau
Berber Piet	UMC Radboud
Thierry Pieters	Cliniques Universitaires Saint-Luc
Joanna Pikiel	Szpitala Pomorskie Sp. z o.o.

Luis Antonio Pires	Instituto de Oncologia de Sorocaba Dr. Gilson Delgado
Albert Pirmagomedov	City Clinical Hospital #1
Renaud Poncin	Clinique Saint Pierre
Dmitry Ponomarenko	Irkutsk Regional Oncology Dispensary
Elvire Pons-Tostivint	CHU Nantes
Sanjay Popat	The Royal Marsden NHS Trust
Kumar Prabhash	Tata Memorial Hospital
Naiyarat Prasongsook	Phramongkutklao Hospital and Medical College
Claudia Proto	Fondazione IRCCS Istituto Nazionale dei Tumori
Mariano Provencio	Hospital Universotario Puerta de Hierro Majadahonda
Dmytro Prygara	Municipal Oncology Centre of Uzhgorod Central Municipal Clinical Hospital
Rodryg Ramlau	Private Specialist Hospitals, MedPolonia
Niels Reinmuth	Asklepios Klinik Gauting GmbH, Asklepios Fachkliniken München-Gauting
Alina Rekuta	Medical Center “Ok Clinic,” International Institute of Clinical Studies LLC
Gary Richardson	Cabrini Medical Centre
Danilo Rocco	Ospedale Monaldi
Jeronimo Rafael Rodriguez	Médica Sur
Steven Rousey	Minnesota Oncology Hematology, P.A.
Paola Claudia Sacco	Ospedale San Giuseppe Moscati di Avellino
Dina Sakaeva	Bashkir State Medical University
Armando Santoro	Istituto Clinico Humanitas
Victor Santos	Ministerio da Saude, Instituto Nacional do Cancer

Wolfgang Schuette	Städtisches Krankenhaus Martha-Maria Halle-Dölau gGmbH
Martin Schuler	Universitaetsklinikum Essen
Tatiana Semiglazova	N.N. Petrov Research Institute Of Oncology
Mehmet Ali Nahit Sendur	Ankara Bilkent City Hospital
Hong Shen	The Second Affiliated Hospital of Zhejiang University College of Medicine
Yuki Shinno	National Cancer Center Hospital
David Robert Spigel	Sarah Cannon Research Institute
Alexander I. Spira	Virginia Cancer Specialists
Sulene Cunha Souza Oliveira	Liga Norte Riograndense Contra O Cancer
Erich Stoelben	Kliniken der Stadt Koeln gGmbH
Daniil Stroyakovskiy	Moscow City Oncology Hospital № 62
Shawgi Sukumaran	Flinders Medical Centre
Ping Sun	Yantai Yuhuangding Hospital
Patrapim Sunpaweravong	Songklanagarind Hospital, Prince of Songkla University
Barna Szima	Markusovszky Egyetemi Oktatókórház
Hiroaki Takeoka	National Hospital Organization Kyushu Medical Center
Hiroshi Tanaka	Niigata Cancer Center Hospital
Kejing Tang	The First Affiliated Hospital, Sun Yat-sen University
Muthukkumaran Thiagarajan	Hospital Kuala Lumpur
Michael Thomas	Thoraxklinik am Universitätsklinikum Heidelberg
Katsuki Tiscoski	Irmandade Santa Casa de Misericórdia de Porto Alegre
Anton Tishchenko	Tambov Regional Oncology Clinical Dispansary
Pascale Tomasini	Hopital Nord
Damien Urban	Chaim Sheba Medical Center

László Urbán	Mátrai Gyógyintézet-Bronchológia
Evgeny Usynin	Tomsk Cancer Research Institute
James Uyeki	Texas Oncology P.A.
Juan Carlos Vázquez Limón	Hospital Civil de Guadalajara Fray Antonio Alcalde
Rémi Veillon	CHU Bordeaux
Alain Vergnenegre	CHU de Limoges
Bo Wang	Willamette Valley Cancer Institute and Research Center
Chin-Chou Wang	Chang Gung Medical Foundation
Jialei Wang	Fudan University Shanghai Cancer Center
Yu-Feng Wei	E-DA Hospital
Paul Wheatley-Price	The Ottawa Hospital Cancer Centre
Ka Yeung Mark Wong	Westmead Hospital
Lin Wu	Hunan Cancer Hospital
Hailin Xiong	Huizhou Municipal Central Hospital
James Chih-Hsin Yang	National Taiwan University Hospital
Yu Yao	The First Affiliated Hospital of Xi'an Jiaotong University
Ozan Yazici	Gazi University Hospital
Akihiro Yoshii	National Hospital Organization Shibukawa Medical Center
Hiroshige Yoshioka	Kansai Medical University Hospital
Yiping Zhang	Zhejiang Cancer Hospital
Jun Zhao	Beijing Cancer Hospital
Yanqiu Zhao	Henan Cancer Hospital
Diansheng Zhong	Tianjin Medical University General Hospital
Lin Zhou	West China Hospital, Sichuan University
Xiangdong Zhou	Southwest Hospital

Ruslan Zukov	Krasnoyarsk Regional Oncology Dispensary
Bogdan Zurawski	Centrum Onkologii im. Prof. F. Lukaszczyka

SUPPLEMENTARY METHODS

Additional Inclusion and Exclusion Criteria

Participants must have had at least 1 measurable lesion, according to Response Evaluation Criteria in Solid Tumors version 1.1) that had not been previously irradiated. Measurable lesions should not have been biopsied during screening; if only 1 non-irradiated measurable lesion exists, it may have undergone a diagnostic biopsy and be acceptable as a target lesion, provided the baseline tumor assessment scans are performed at least 14 days after the biopsy. Participants must have had an Eastern Cooperative Oncology Group performance status of 0 or 1.

Participants must not have received any prior epidermal growth factor receptor–tyrosine kinase inhibitor or systemic treatment at any time for locally advanced Stage III or metastatic Stage IV disease (adjuvant or neoadjuvant therapy for Stage I or II disease was allowed if administered more than 12 months prior to the development of locally advanced or metastatic disease).

Participants with asymptomatic or previously treated and stable brain metastases were allowed in this study. Participants who had received definitive radiation or surgical treatment for symptomatic or unstable brain metastases and been clinically stable and asymptomatic for at least 2 weeks before randomization were eligible, provided they have been either off corticosteroid treatment or are receiving low-dose corticosteroid treatment (≤ 10 mg/day prednisone or equivalent) for at least 2 weeks prior to randomization. Participants were not eligible if they have received an investigational medication within 12 months before randomization or are currently enrolled in an investigational study. Participants must not have had a concurrent or prior malignancy other than the disease under study or had an uncontrolled illness, with exceptions (please see the protocol for more details, available at NEJM.org).

Prophylactic Anticoagulation

All patients receiving amivantamab-lazertinib are recommended to receive prophylactic-dose anticoagulation as per local guidelines during the first 4 months of treatment.

This recommendation was implemented in August 2022, when the majority of the patients were past the initial 4 months of treatment. In ongoing studies, the uptake of prophylactic anticoagulation is expected to be higher, which will allow for the prospective evaluation of this mitigation strategy.

Statistical Analysis

One futility analysis and one interim analysis were conducted as planned when approximately 120 and 280 progression-free survival events occurred, respectively, from the amivantamab-lazertinib and osimertinib groups combined.

At the time of the interim progression-free survival analysis, there were 321 progression-free survival events by blinded independent central review observed from the amivantamab-lazertinib and osimertinib groups combined. After reviewing the data, the independent data monitoring committee notified the sponsor committee that the efficacy stopping criteria had been met for progression-free survival (hazard ratio for disease progression or death, 0.75; 95% CI, 0.60 to 0.93; $P=0.0097$ [less than the 2-sided significance level criterion of 0.0159 based on the O'Brien-Fleming stopping boundary]), in favor of the amivantamab-lazertinib group. At the time of the clinical cutoff for the interim progression-free survival analysis, the median duration of follow-up was only 15.1 months, meaning data observed at the interim analysis might not have reflected the true treatment effect. Accordingly, to better inform the benefit-risk profile, the sponsor committee decided to continue the study and maintain the blinding until the next protocol-specified progression-free survival analysis (the final progression-free survival analysis

based on 450 progression-free survival events from the amivantamab-lazertinib and osimertinib groups combined). This decision was endorsed by the independent data monitoring committee and by regulators, and the study continued as planned with the sponsor study team and investigators remaining blinded.

Overall survival was to be analyzed with an overall type I error rate of 0.05 given the statistical significance in PFS was achieved. One interim analysis of overall survival was planned and occurred at the time of final analysis of PFS. With 214 deaths in the amivantamab-lazertinib and osimertinib groups combined, based on the O'Brien Fleming alpha spending approach, a 2-sided alpha of 0.005 was allocated to the interim analysis (55% maturity). The final analysis of overall survival is planned to occur when approximately 490 total deaths (390 in the amivantamab-lazertinib and osimertinib groups combined) are anticipated. Overall survival was analyzed using the same methodology and model as for the analysis of progression-free survival.

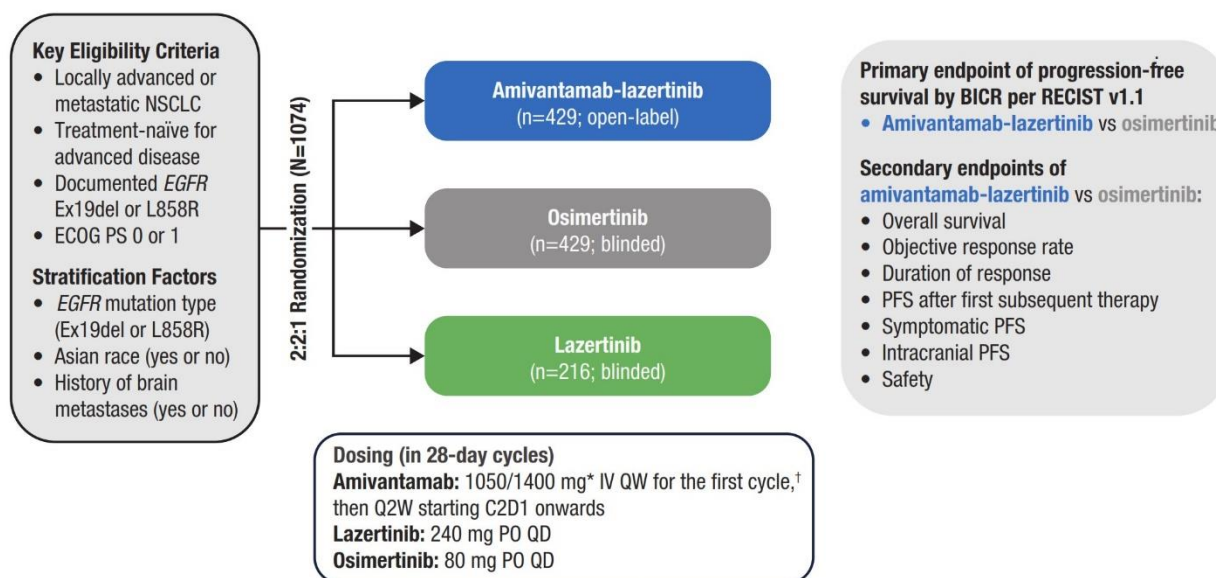
A comparison of amivantamab-lazertinib versus lazertinib in progression-free survival and overall survival was performed to assess the contribution of amivantamab to the activity of the combination. There was no formal hypothesis testing; summary statistics were used.

The proportional hazards assumption was examined by plotting $\log(-\log[\text{estimated survival distribution function}])$ against $\log(\text{survival time})$. The assumption appeared to be satisfied for all analyses that used the Cox model.

SUPPLEMENTARY RESULTS

Supplementary Figures

Figure S1. MARIPOSA Study Design.



C denotes cycle, BICR blinded independent central review, D day, ECOG PS Eastern Cooperative Oncology Group performance status, EGFR epidermal growth factor receptor, Ex19del Exon 19 deletion, IV intravenous, L858R Exon 21 L858R, NSCLC, non-small cell lung cancer, PFS progression-free survival, PO oral, Q2W every 2 weeks, QD daily, QW weekly, and RECIST Response Evaluation Criteria in Solid Tumors.

*1400-mg doses for patients with body weight 80 kg or greater.

[†]Cycle 1, day 1/2 (split dose), 8, 15, and 22.

Figure S2. Consolidated Standards of Reporting Trials (CONSORT) Diagram of Patient Disposition.

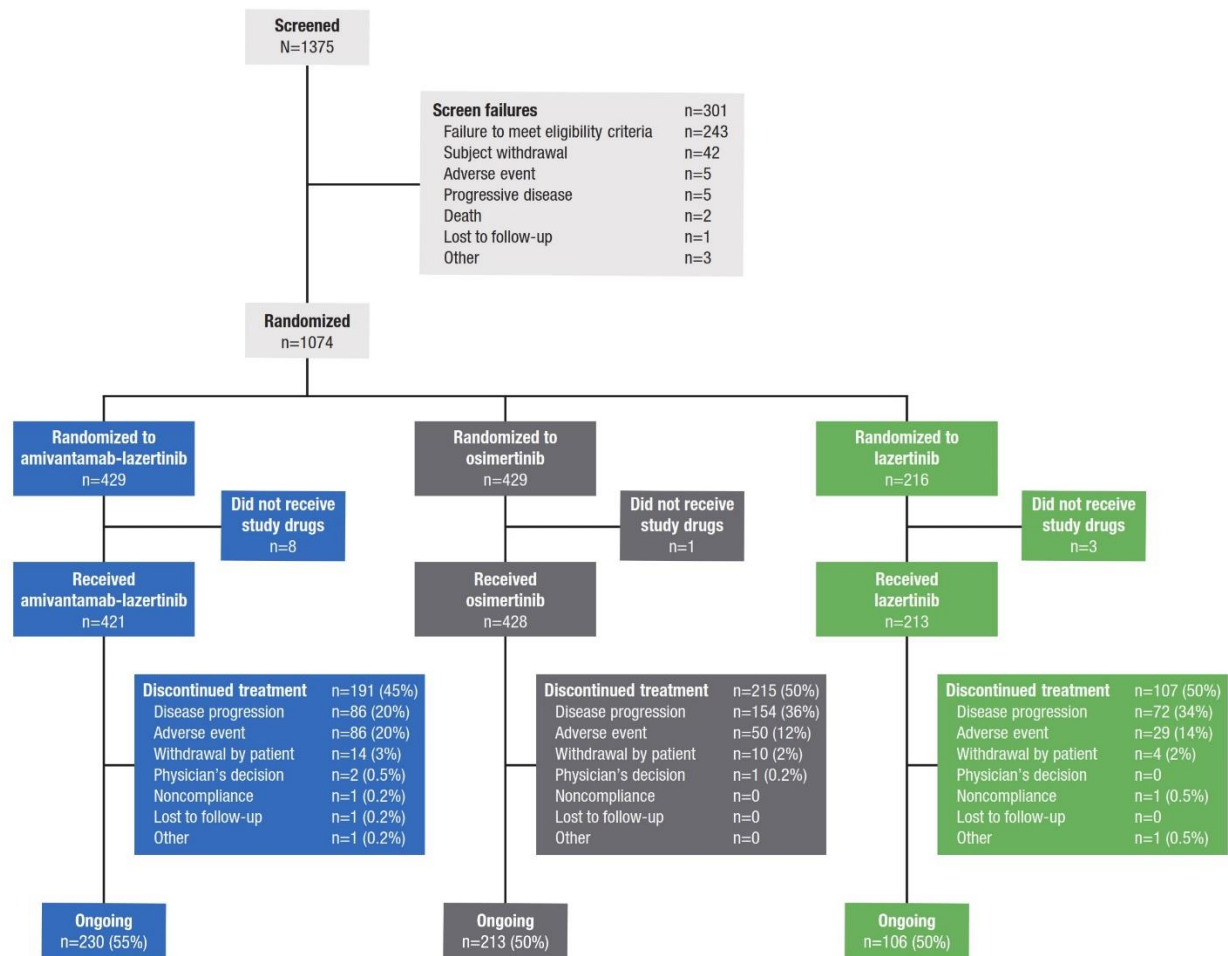


Figure S3. Progression-free Survival by Blinded Independent Central Review for *EGFR*-mutation Subgroups.

Shown are Kaplan-Meier estimates of progression-free survival as assessed by blinded independent central review in patients with *EGFR* Exon 19 deletions (Panel A) and in patients with *EGFR* Exon 21 L858R (Panel B). The efficacy population included all patients who had undergone randomization. In both panels, the hazard ratio for disease progression or death was obtained from an unstratified proportional hazards model; tick marks indicate censoring of data, and the shaded area indicate 95% confidence bands. The 95% CI widths for the hazard ratio have not been adjusted for multiplicity and should not be used to infer definitive treatment effects. CI denotes confidence interval and EGFR epidermal growth factor receptor.

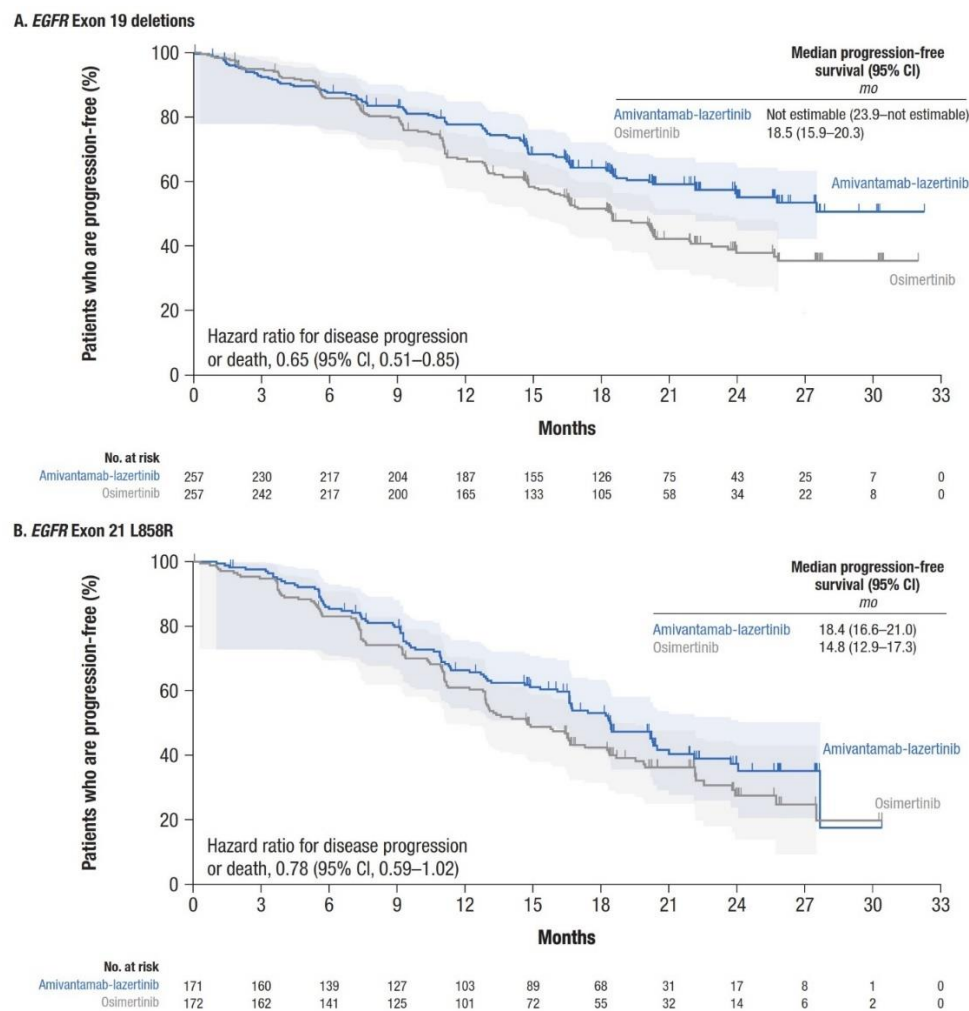


Figure S4. Progression-free Survival by Blinded Independent Central Review for Asian/non-Asian Subgroups.

Shown are Kaplan-Meier estimates of progression-free survival as assessed by blinded independent central review in patients who are of Asian race (Panel A) and in patients who are of non-Asian race (Panel B). The efficacy population included all patients who had undergone randomization. In both panels, the hazard ratio for disease progression or death was obtained from an unstratified proportional hazards model; tick marks indicate censoring of data, and the shaded area indicate 95% confidence bands. 95% CI widths for hazard ratio have not been adjusted for multiplicity and should not be used to infer definitive treatment effects. CI denotes confidence interval.

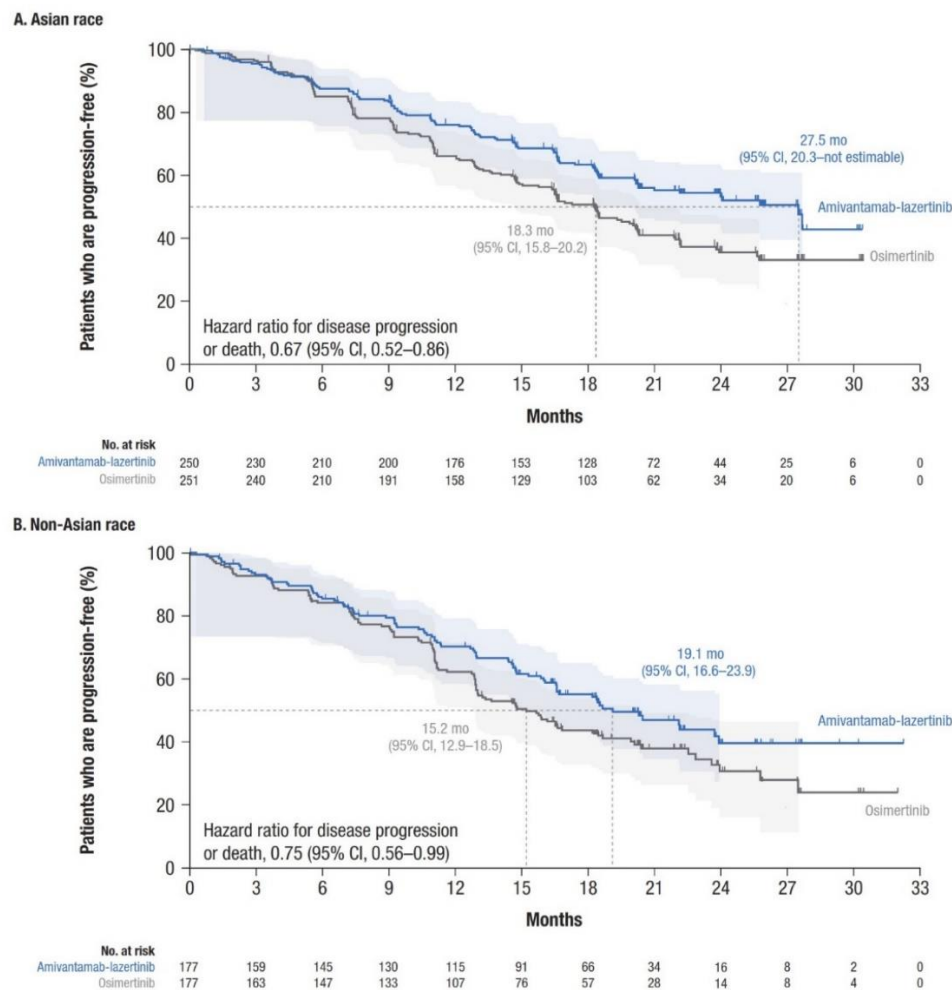


Figure S5. Progression-free Survival by Blinded Independent Central Review for History of Brain Metastases Subgroups.

Shown are Kaplan-Meier estimates of progression-free survival as assessed by blinded independent central review in patients with a history of brain metastases (Panel A) and in patients without a history of brain metastases (Panel B). The efficacy population included all patients who had undergone randomization. In both panels, the hazard ratio for disease progression or death was obtained from an unstratified proportional hazards model; tick marks indicate censoring of data, and the shaded area indicate 95% confidence bands. The 95% CI widths for hazard ratio have not been adjusted for multiplicity and should not be used to infer definitive treatment effects. CI denotes confidence interval.

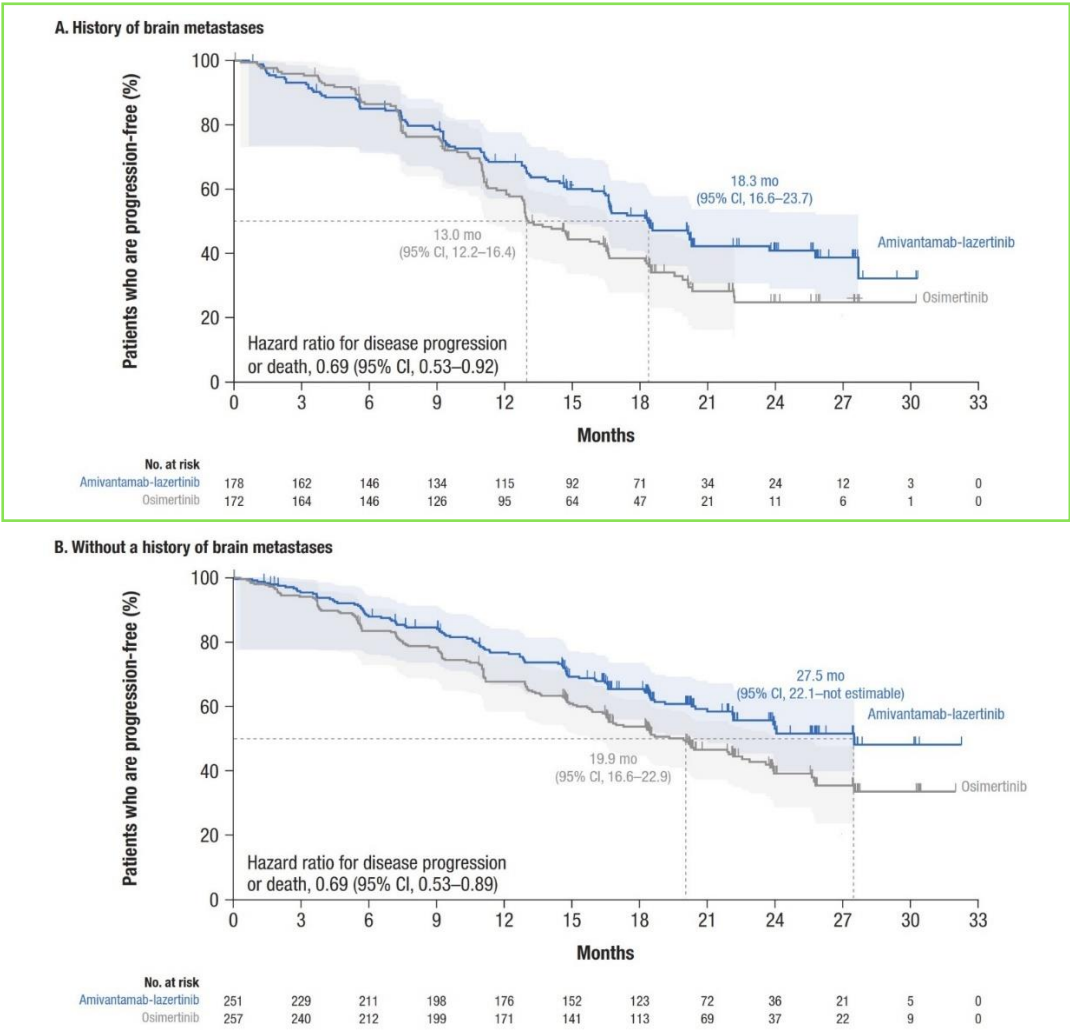


Figure S6. Extracranial Progression-free Survival by Blinded Independent Central Review.

Shown are Kaplan-Meier estimates of extracranial progression-free survival as assessed by blinded independent central review in the efficacy population. The efficacy population included all patients who had undergone randomization. Extracranial progression-free survival was defined as the time from randomization to disease progression (detected by extracranial scans) or death. If first progression was detected solely in the CNS, these patients were censored at the time of CNS disease progression. The dashed lines indicate the median extracranial progression-free survival in the two groups; tick marks indicate censoring of data, and the shaded area indicate 95% confidence bands. The 95% CI widths for hazard ratio have not been adjusted for multiplicity and should not be used to infer definitive treatment effects. CI denotes confidence interval and CNS central nervous system.

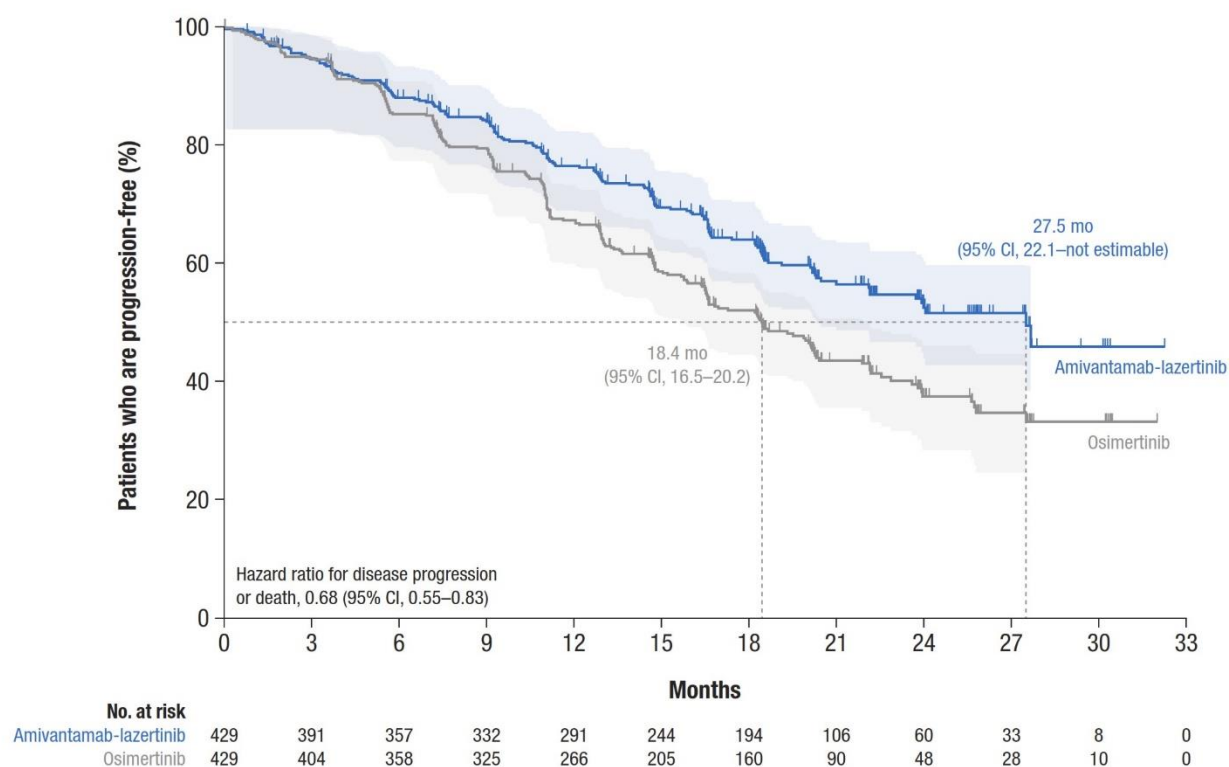


Figure S7. Best Percent Change From Baseline in Target Lesions (Waterfall Plots).

Shown are waterfall plots of the best percent change from baseline in target lesions in the amivantamab-lazertinib group (Panel A) and in the osimertinib group (Panel B). The number of patients with measurable disease at baseline was 421 in amivantamab-lazertinib group and 414 in the osimertinib group, as determined by blinded independent central review. Target lesions were measured as the sum of the diameters.

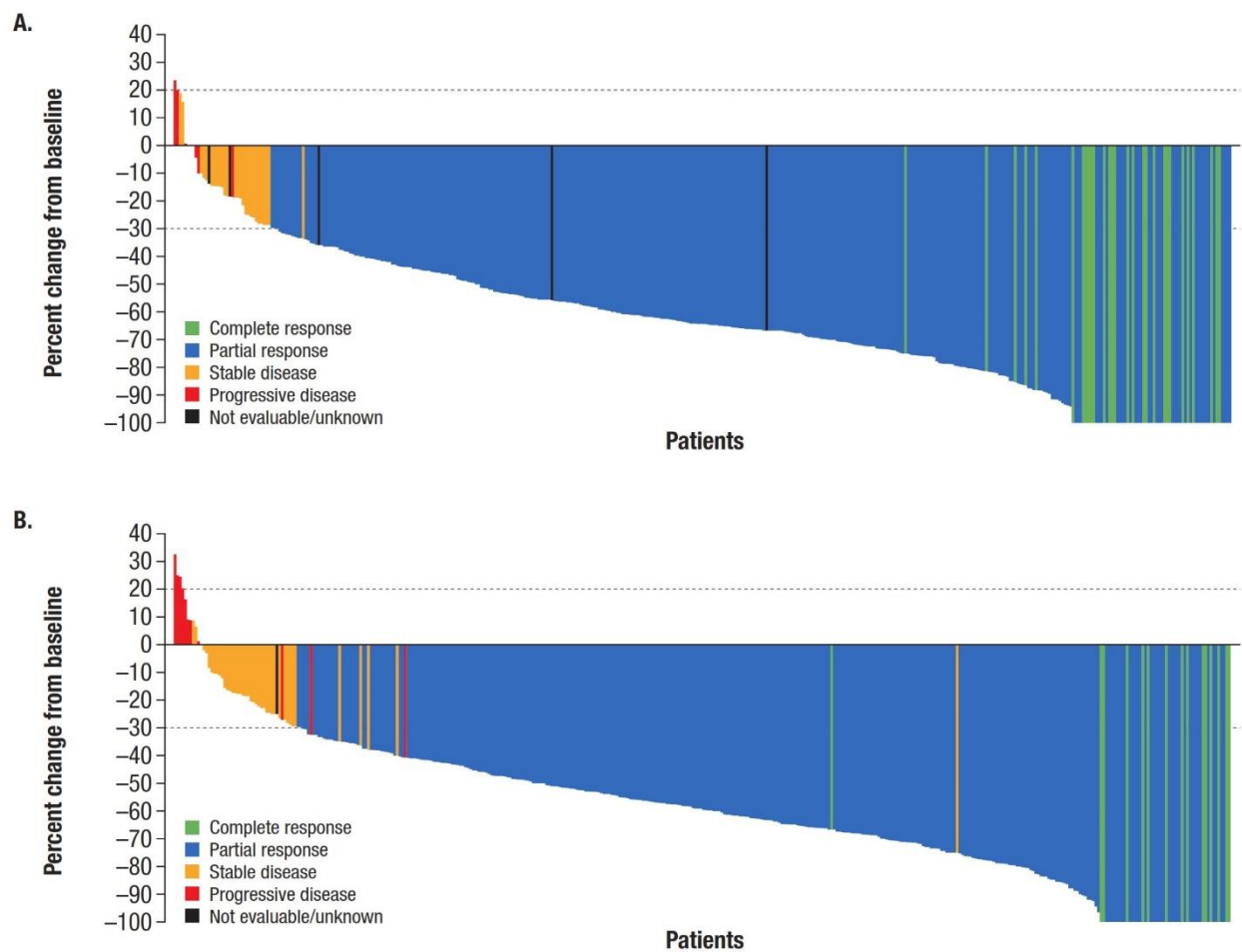


Figure S8. Response Duration.

Shown is a Kaplan-Meier estimate of response duration among confirmed responders in the efficacy population. The efficacy population included all patients who had undergone randomization. Included in this analysis were the 336 confirmed responders (out of the 421 patients with measurable disease at baseline by blinded independent central review) in the amivantamab-lazertinib group and the 314 confirmed responders (out of 414 patients) in the osimertinib group. Tick marks indicate censoring of data, and the shaded area indicate 95% confidence bands. CI denotes confidence interval.

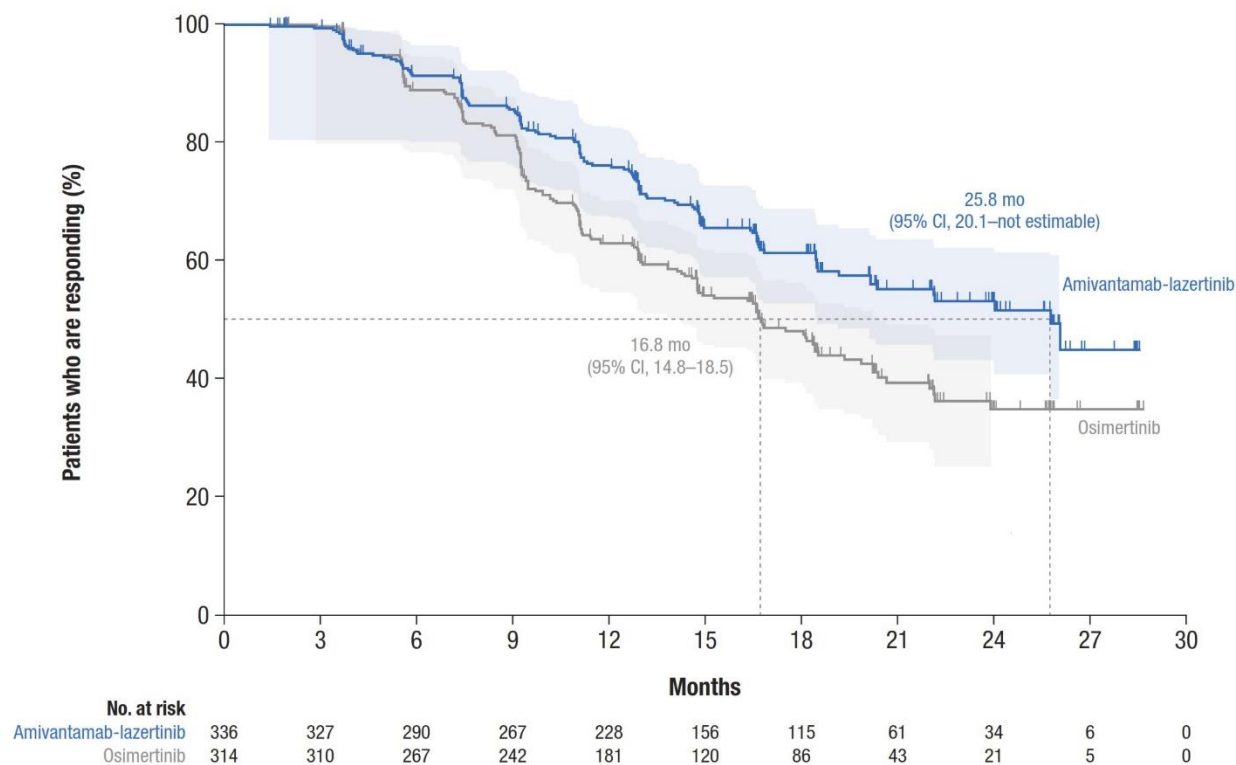


Figure S9. Time to Treatment Discontinuation.

Shown is a Kaplan-Meier estimate of time to treatment discontinuation in the efficacy population. The efficacy population included all patients who had undergone randomization. Tick marks indicate censoring of data, and the shaded area indicate 95% confidence bands. The 95% CI widths for hazard ratio have not been adjusted for multiplicity and should not be used to infer definitive treatment effects. CI denotes confidence interval.

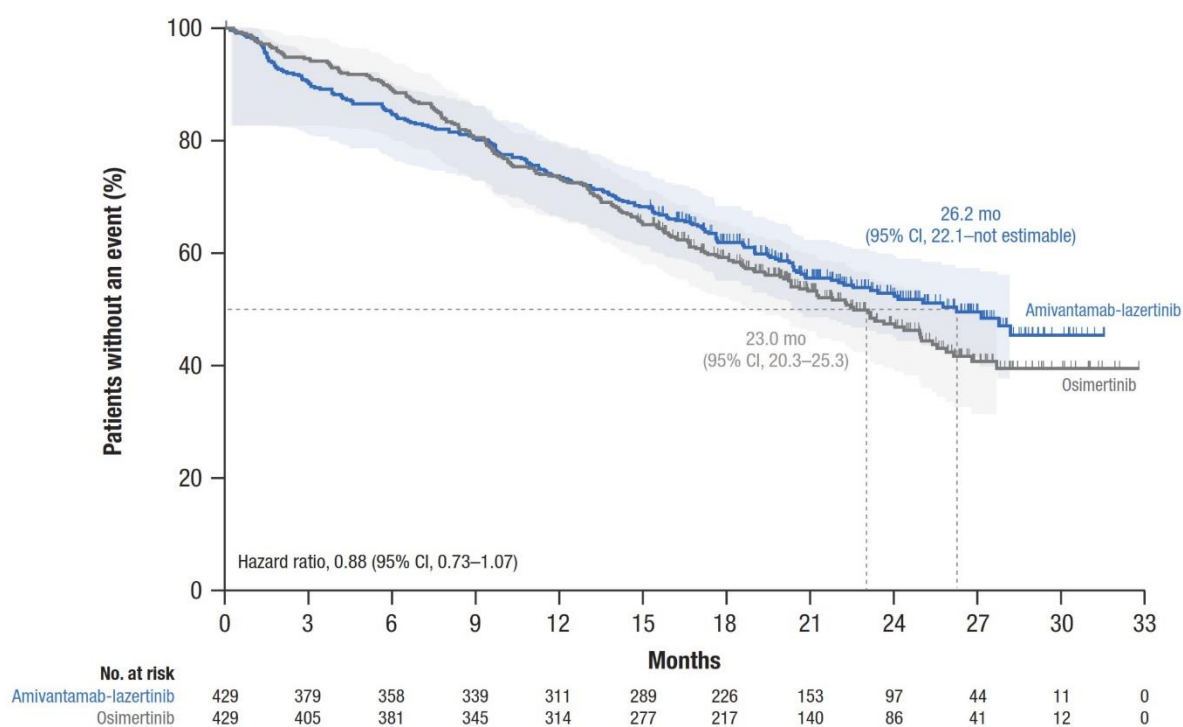


Figure S10. Time to Subsequent Therapy.

Shown are Kaplan-Meier estimates of the time to subsequent therapy in the efficacy population. The efficacy population included all patients who had undergone randomization. Tick marks indicate censoring of data, and the shaded area indicate 95% confidence bands. The 95% CI widths for hazard ratio have not been adjusted for multiplicity and should not be used to infer definitive treatment effects. CI denotes confidence interval.

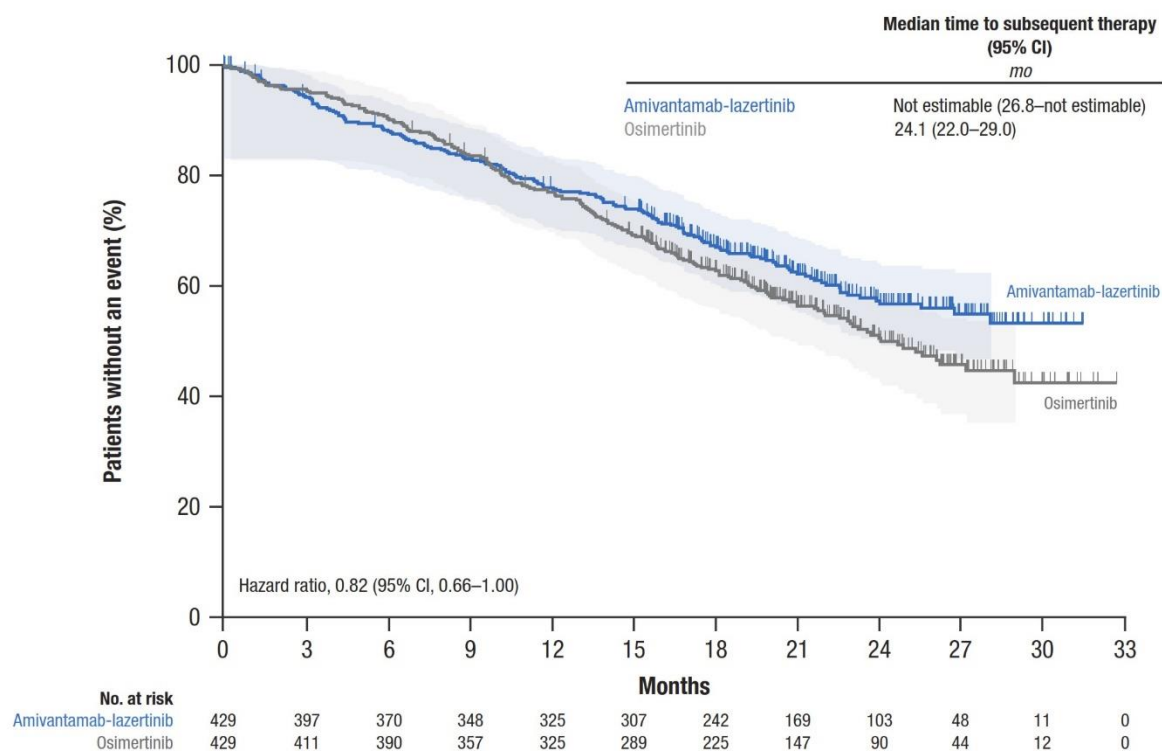
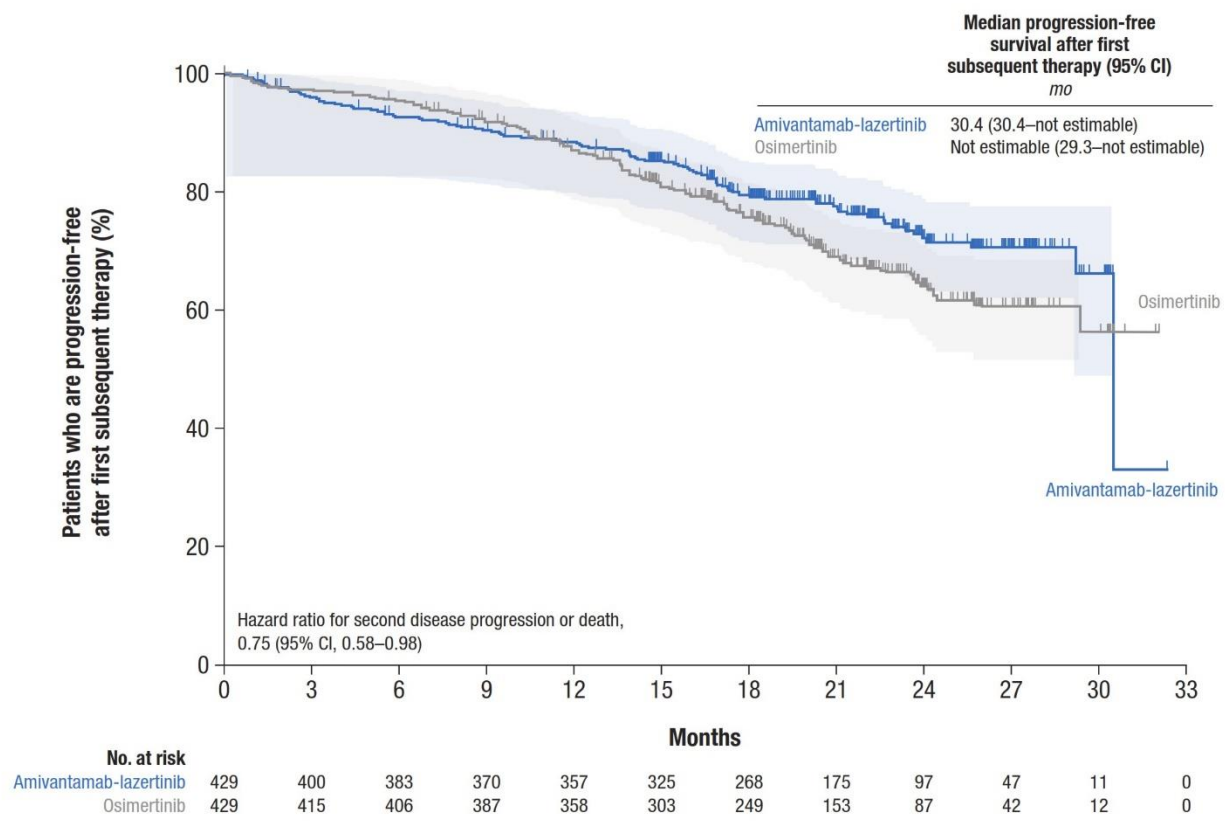


Figure S11. Progression-free Survival After First Subsequent Therapy.

Shown is a Kaplan-Meier estimate of progression-free survival after first subsequent therapy, which was defined from the time from randomization until the date of objective disease progression (by investigator) or death after initiation of subsequent anticancer systemic therapy, whichever occurs first. The efficacy population included all patients who had undergone randomization. Tick marks indicate censoring of data, and the shaded area indicate 95% confidence bands. The 95% CI widths for hazard ratio have not been adjusted for multiplicity and should not be used to infer definitive treatment effects. CI denotes confidence interval.



Supplementary Tables

Table S1. Representativeness of Study Participants

Category	Details
Disease under investigation	<i>EGFR</i> -mutated (Ex19del or Exon 21 L858R) NSCLC
Special considerations related to:	
Sex	<i>EGFR</i> -mutated NSCLC affects more females (43.7%) than males (24.0%) ¹
Age	The median age from a meta-analysis* of 456 NSCLC studies was 63 years ¹
Race or ethnic group	Overall, the highest prevalence of <i>EGFR</i> -mutated NSCLC was in Asian populations (38.8%), followed by 17.4% in Caucasians, 17.2% in African Americans, and 27.0% in mixed populations. ¹ In a prospective analysis of 1482 Asian patients, the <i>EGFR</i> mutation frequency with advanced NSCLC (adenocarcinoma histology) was 51.4%, where the most commonly detected mutations were Ex19del (alone: 22.1%; in combination with others: 24.3%) and L858R (alone: 20.9%; in combination with others: 22.9%) ²
Geography	Similar to the estimates by race, the highest prevalence of <i>EGFR</i> -mutated NSCLC was in the Asia-Pacific region (point estimates vary from 38.4%* to 47%†) and the lowest was in Oceania (12%†). ^{1,3} Based on model estimates,‡ Ex19del is the most prominent in South America (66.8%) and the least

	prominent in Oceania (40.3%), while L858R is most prominent in Asia (41.1%) and least prominent in South America (27.7%) ⁴
Other considerations	<i>EGFR</i> -mutated NSCLC is more commonly observed in patients who are never smokers (49.3% vs 21.5% for those with history of smoking*) and with the adenocarcinoma histology type (38.0% vs 11.7% for non-adenocarcinoma*). ¹ The prevalence of <i>EGFR</i> Ex19del and L858R are similar among NSCLC disease stages (I, 34.0%; II, 29.9%; III, 33.8%; IV, 37.5%)*
Overall representativeness of this trial	The demographic and epidemiologic characteristics of the MARIPOSA patient population were generally consistent with the current data for patients with <i>EGFR</i> -mutated NSCLC. ¹ There were more females than males, most patients did not have a history of smoking, and the highest prevalence of <i>EGFR</i> -mutated NSCLC was observed in Asian and White patients. Some racial groups were underrepresented in this trial, such as patients who are Black or African American.

EGFR denotes epidermal growth factor receptor, Ex19del Exon 19 deletion, and NSCLC non-small cell lung cancer.

*Obtained from a systematic review and meta-analysis of 456 studies of patients with NSCLC.

†Pooled prevalence of *EGFR* mutations in patients with NSCLC; data obtained from a systematic review of 87 studies.

‡Values obtained from linear mixed-effects models fitted to *EGFR*-mutation endpoints using logistic transformation, assuming a binomial distribution, of 74 studies with *EGFR* Ex19del and L858R NSCLC.

Table S2. Demographics and Baseline Disease Characteristics of the Lazertinib Monotherapy

Group

Characteristic	Lazertinib (n=216)
Age	
Median (range) — years	63 (31–87)
Category — no. (%)	
<65 years	119 (55)
≥65 to <75 years	79 (37)
≥75 years	18 (8)
Sex — no. (%)	
Female	136 (63)
Male	80 (37)
Race* — no. (%)	
Asian	128 (59)
White	79 (37)
American Indian or Alaska Native	4 (2)
Black or African American	4 (2)
Native Hawaiian or Pacific Islander	0
Multiple	0
Unknown	1 (<1)
Body weight	
Median (range) — kg	60.5 (41–118)
Category — no. (%)	
<80 kg	197 (91)
≥80 kg	19 (9)
Eastern Cooperative Oncology Group performance status — no. (%)	

0	76 (35)
1	140 (65)
History of smoking — no. (%)	
No	143 (66)
Yes	73 (34)
Time from initial diagnosis, median (range) — months	
	1.3 (0.2–197.3)
Time from metastatic diagnosis, median (range) — months	
	1.2 (0.2–9.2)
Histologic type — no. (%)	
Adenocarcinoma	212 (98)
Large cell carcinoma	0
Squamous cell carcinoma	2 (1)
Other†	2 (1)
Not reported	0
History of brain metastases — no. (%)	
	86 (40)
Type of <i>EGFR</i> mutation — no. (%)	
Exon 19 deletion	131 (61)
Exon 21 L858R	85 (39)

EGFR denotes epidermal growth factor receptor.

*Race or ethnic group was reported by the patients.

†Other histologic types included: adenocarcinoma and squamous cell carcinoma, lepidic adenocarcinoma, non-small cell carcinoma, pleomorphic carcinoma, and unknown.

Table S3. First Subsequent Systemic Therapy*

Therapy — no. of patients (%)	Amivantamab-	
	lazertinib (n=429)	Osimertinib (n=429)
With investigator-assessed disease progression and discontinued all randomized treatment	116	171
Any first subsequent therapy	78 (67)	124 (73)
EGFR-targeted or TKI-based regimen	37 (32)	38 (22)
EGFR-TKI only	31 (27)	29 (17)
Osimertinib	19 (16)	19 (11)
Gefitinib	4 (3)	2 (1)
Furmonertinib	4 (3)	1 (1)
Aumolertinib	2 (2)	2 (1)
Icotinib	1 (1)	1 (1)
Catequentinib	1 (1)	0
Afatinib	0	2 (1)
Erlotinib	0	1 (1)
Afatinib + osimertinib	0	1 (1)
EGFR-targeted or TKI-based combination	6 (5)	9 (5)
Osimertinib + carboplatin + pemetrexed	1 (1)	1 (1)
Osimertinib + cisplatin + pemetrexed	1 (1)	1 (1)

Afatinib + bevacizumab + carboplatin + pemetrexed	1 (1)	0
Aumolertinib + bevacizumab	1 (1)	0
Erlotinib + carboplatin + pemetrexed	1 (1)	0
Osimertinib + bevacizumab + carboplatin + pemetrexed	1 (1)	0
Aumolertinib + HS-10241	0	2 (1)
Amivantamab + lazertinib	0	1 (1)
Aumolertinib + carboplatin + etoposide	0	1 (1)
Icotinib + carboplatin + etoposide	0	1 (1)
Osimertinib + capmatinib	0	1 (1)
Osimertinib + tepotinib	0	1 (1)
Chemotherapy or immunotherapy-based regimens	40 (34)	81 (47)
Chemotherapy alone	30 (26)	50 (29)
Carboplatin + pemetrexed	16 (14)	26 (15)
Carboplatin + paclitaxel	6 (5)	10 (6)
Carboplatin + etoposide	2 (2)	0
Cisplatin + etoposide	2 (2)	0
Cisplatin + pemetrexed	1 (1)	9 (5)
Pemetrexed	1 (1)	1 (1)
Carboplatin	1 (1)	0
Carboplatin + vinorelbine	1 (1)	0

Carboplatin + cisplatin + gemcitabine + paclitaxel	0	1 (1)
Carboplatin + docetaxel	0	1 (1)
Cisplatin + docetaxel	0	1 (1)
Vinorelbine	0	1 (1)
Chemotherapy plus VEGF inhibitors	3 (3)	10 (6)
Carboplatin + pemetrexed + bevacizumab	2 (2)	3 (2)
Carboplatin + paclitaxel + bevacizumab	1 (1)	1 (1)
Cisplatin + pemetrexed + bevacizumab	0	3 (2)
Pemetrexed + bevacizumab	0	2 (1)
Carboplatin + cisplatin + pemetrexed + bevacizumab	0	1 (1)
Chemotherapy plus immunotherapy	1 (1)	8 (5)
Cisplatin + paclitaxel + serplulimab	1 (1)	0
Carboplatin + pemetrexed + pembrolizumab	0	4 (2)
Carboplatin + pemetrexed + camrelizumab	0	2 (1)
Carboplatin + paclitaxel + atezolizumab	0	1 (1)
Carboplatin + durvalumab + etoposide	0	1 (1)
Chemotherapy combined with VEGF inhibitors and immunotherapy	5 (4)	13 (8)
Carboplatin + paclitaxel + bevacizumab + atezolizumab	3 (3)	9 (5)

Carboplatin + pemetrexed + bevacizumab + sintilimab	1 (1)	1 (1)
Cisplatin + paclitaxel + bevacizumab + atezolizumab	1 (1)	0
Carboplatin + pemetrexed + bevacizumab + atezolizumab	0	1 (1)
Cisplatin + pemetrexed + bevacizumab + sintilimab	0	1 (1)
Pemetrexed + bevacizumab + atezolizumab	0	1 (1)
VEGF inhibitor only	1 (1)	0
Bevacizumab	1 (1)	0
Other	1 (1)	5 (3)
Amivantamab monotherapy	0	1 (1)
Antibody drug conjugate	0	1 (1)
Herbal	1 (1)	1 (1)
Antineoplastic agents	0	2 (1)

EGFR denotes epidermal growth factor receptor, TKI tyrosine kinase inhibitor, and VEGF vascular endothelial growth factor.

*The efficacy population included all the patients who had undergone randomization (n=429 for amivantamab-lazertinib and n=429 for osimertinib); shown here are percentages among the 116 patients in the amivantamab-lazertinib group and 171 patients in the osimertinib group who had investigator-assessed disease progression on and discontinued their randomized treatment.

Table S4. Contribution of Amivantamab on Progression-free Survival, Overall Survival, and Objective Response*

Comparison of amivantamab-lazertinib versus lazertinib (95% CI†)	
Progression-free-survival	Hazard ratio, 0.72 (0.57–0.90)
Overall survival	Hazard ratio, 0.82 (0.59–1.14)
Objective response	Odds ratio, 1.31 (0.83–2.06)§

CI denotes confidence interval.

*The efficacy population included all the patients who had undergone randomization (n=429 for amivantamab-lazertinib and n=216 for lazertinib).

†The 95% CI widths for hazard ratio and odds ratio have not been adjusted for multiplicity and cannot be used to infer definitive treatment effects

‡The hazard ratios were obtained from a stratified proportional hazards model, stratified by *EGFR* mutation type, Asian race, and history of brain metastasis.

§The objective response (complete or partial response) was assessed by blinded independent central review. Included in the analysis were 421 patients with measurable disease at baseline in the amivantamab-lazertinib group and 214 patients with measurable disease at baseline in the lazertinib group. The odds ratio is from a logistic regression model stratified by *EGFR* mutation type, Asian race, and history of brain metastasis.

Table S5. Response Efficacy Endpoints*

Endpoint	Amivantamab-lazertinib (n=429)	Osimertinib (n=429)
Objective response†		
Patients (95% CI) including all responders — %	86 (83–89)	85 (81–88)
Patients (95% CI) including only confirmed responders — %	80 (76–84)	76 (71–80)
Best overall response† — no. (%)		
Complete response‡	29 (7)	15 (4)
Partial response‡	334 (79)	335 (81)
Stable disease	30 (7)	42 (10)
Progressive disease	7 (2)	11 (3)
Not evaluable	21 (5)	11 (3)
Response duration†		
Median (95% CI) among all responders — mo	24.0 (18.5–NE)	14.9 (12.9–17.5)
Median (95% CI) among confirmed responders — mo	25.8 (20.1–NE)	16.8 (14.8–18.5)

CI denotes confidence interval and NE not estimable.

*The efficacy population included all the patients who had undergone randomization.

†The objective response (complete or partial response) and response duration were assessed by blinded independent central review. Included in the analysis were 421 patients with measurable disease at baseline in the amivantamab-lazertinib group and 414 in the osimertinib group.

‡Includes all responders.

Table S6. Adverse Events Summary for the Lazertinib Monotherapy Group*

Event — no. (%)	Lazertinib (n=213)
Any event	213 (100)
Grade ≥ 3	97 (46)
Any serious event	75 (35)
Any event resulting in death	12 (6)
Any event leading to:	
Interruptions of any study agent	92 (43)
Reductions of any study agent	27 (13)
Discontinuations of any study agent	28 (13)

*The safety population included all the patients who had undergone randomization and received at least one dose of any trial treatment.

Table S7. Serious Adverse Events Occurring in at Least 1% of Patients*

Event† — no. (%)	Amivantamab-lazertinib (n=421)	Osimertinib (n=428)
Pulmonary embolism	26 (6)	10 (2)
Pneumonia	17 (4)	21 (5)
Deep vein thrombosis	12 (3)	2 (<1)
COVID-19	10 (2)	10 (2)
Pleural effusion	9 (2)	17 (4)
Infusion-related reaction	9 (2)	0
Alanine aminotransferase increased	8 (2)	6 (1)
Pneumonitis	7 (2)	8 (2)
Rash	7 (2)	0
Respiratory failure	6 (1)	2 (<1)
Interstitial lung disease	5 (1)	5 (1)
Hyponatremia	5 (1)	4 (1)
Hypoalbuminemia	5 (1)	0
Dyspnea	4 (1)	11 (3)
Pericardial effusion	4 (1)	5 (1)
Cardiac failure	1 (<1)	5 (1)

*The safety population included all the patients who had undergone randomization and received at least one dose of any trial treatment.

†Events in this category are listed according to decreasing incidence in the amivantamab-lazertinib group.

Table S8. Incidence, Time to Onset, and Duration of Select Adverse Events of Interest*

Event — no. (%)	Amivantamab- lazertinib (n=421)	Osimertinib (n=428)
Rash†		
No. of patients (%)	373 (89)	210 (49)
No. of days to first onset — median (range)	14 (1–556)	29 (1–799)
No. of events, any grade	1447	238
Duration of event – median no. of days (range)	31 (1–663)	51 (1–625)
Venous thromboembolism‡		
No. of patients (%)	157 (37)	39 (9)
No. of days to first onset — median (range)	84 (6–777)	194 (10–675)
Within first 4 months§ — no. of patients (%)	97 of 157 (62)	13 of 39 (33)
Interstitial lung disease 		
No. of patients (%)	13 (3)	13 (3)
No. of days to first onset — median (range)	110 (29–443)	115 (9–512)
No. of events, any grade	14	15
Duration of event – median no. of days (range)	18 (1–191)	20 (1–189)

*The safety population included all the patients who had undergone randomization and received at least one dose of any trial treatment.

†Included the following preferred terms: rash, dermatitis acneiform, folliculitis, rash maculopapular, skin lesion, acne, erythema, rash pustular, dermatitis, rash pruritic, rash papular, rash erythematous, rash macular, dermatitis infected, erythema multiforme, papule, drug eruption, rash follicular, rash vesicular, skin exfoliation, and epidermolysis.

‡Included the following preferred terms: pulmonary embolism, deep vein thrombosis, venous thrombosis limb, thrombosis, venous thrombosis, superficial vein thrombosis, thrombophlebitis, embolism, embolism venous, jugular vein thrombosis, pulmonary infarction, axillary vein thrombosis, portal vein thrombosis, post thrombotic syndrome, sigmoid sinus thrombosis, superior sagittal sinus thrombosis, vena cava thrombosis, pelvic venous thrombosis, and pulmonary thrombosis.

§Incidence of VTE between 5 to 8 months and after 8 months (starting at month 9) of treatment was 17% and 21% in the amivantamab-lazertinib group, respectively, and 21% and 46% in the osimertinib group, respectively.

||Included the following preferred terms: pneumonitis and interstitial lung disease.

Table S9. Venous Thromboembolism Events and Anticoagulation Use*

Event — no. (%)	Amivantamab- lazertinib (n=421)	Osimertinib (n=428)
Any venous thromboembolism event	157 (37)	39 (9)
Grade 1	5 (1)	0
Grade 2	105 (25)	24 (6)
Grade 3	43 (10)	12 (3)
Grade 4	2 (<1)	1 (<1)
Grade 5	2 (<1)	2 (<1)
Any venous thromboembolism event leading to death	2 (<1)	2 (<1)
Any venous thromboembolism event leading to discontinuation of any agent	12 (3)	2 (<1)
Venous thromboembolism events†		
Pulmonary embolism	73 (17)	20 (5)
Deep vein thrombosis	61 (14)	11 (3)
Venous thrombosis limb	17 (4)	1 (<1)
Thrombosis	9 (2)	1 (<1)
Venous thrombosis	8 (2)	1 (<1)
Superficial vein thrombosis	6 (1)	0
Thrombophlebitis	6 (1)	4 (1)
Embolism	4 (1)	3 (1)

Embolism venous	4 (1)	1 (<1)
Jugular vein thrombosis	3 (1)	0
Pulmonary infarction	2 (<1)	0
Axillary vein thrombosis	1 (<1)	0
Portal vein thrombosis	1 (<1)	0
Post thrombotic syndrome	1 (<1)	0
Sigmoid sinus thrombosis	1 (<1)	0
Superior sagittal sinus thrombosis	1 (<1)	0
Vena cava thrombosis	1 (<1)	0
Pelvic venous thrombosis	0	1 (<1)
Pulmonary thrombosis	0	1 (<1)
Superior vena cava syndrome	0	1 (<1)
Anticoagulation use		
At baseline	21 (5)	23 (5)
Concomitant use	199 (47)	81 (19)
First bleeding event while on anticoagulants	34 (8)	14 (3)
Recurrent venous thromboembolism event while on anticoagulants	8 (2)	0

*The safety population included all the patients who had undergone randomization and received at least one dose of any trial treatment.

†Events in this category are listed according to decreasing incidence in the amivantamab-lazertinib group.

Table S10. Adverse Events Leading to Treatment Interruptions, Reductions, and Discontinuations*

	Amivantamab-lazertinib	Osimertinib
Event — no. (%)	(n=421)	(n=428)
Any event leading to interruptions of any study agent	350 (83)†	165 (39)
Most common events leading to interruptions of any study agent‡		
Rash	104 (25)	4 (1)
Paronychia	91 (22)	4 (1)
COVID-19	63 (15)	36 (8)
Dermatitis acneiform	50 (12)	1 (<1)
Increased alanine aminotransferase	30 (7)	12 (3)
Hypoalbuminemia	25 (6)	0
Increased aspartate aminotransferase	23 (5)	11 (3)
Pulmonary embolism	20 (5)	9 (2)
Peripheral edema	20 (5)	0
Diarrhea	19 (5)	6 (1)
Deep vein thrombosis	19 (5)	2 (<1)
Asthenia	18 (4)	1 (<1)
Pneumonia	14 (3)	18 (4)

Increased γ -glutamyltransferase	13 (3)	5 (1)
Stomatitis	13 (3)	4 (1)
Any event leading to dose reductions of any study agent	249 (59) [†]	23 (5)
Most common events leading to dose reductions of any study agent [§]		
Rash	84 (20)	2 (<1)
Paronychia	80 (19)	1 (<1)
Dermatitis acneiform	38 (9)	0
Hypoalbuminemia	11 (3)	0
Increased alanine aminotransferase	8 (2)	0
Paresthesia	8 (2)	0
Peripheral edema	7 (2)	0
Folliculitis	6 (1)	0
Rash maculo-papular	6 (1)	0
Increased aspartate aminotransferase	5 (1)	0
Mucosal inflammation	5 (1)	0
Neuropathy peripheral	5 (1)	0
Peripheral sensory neuropathy	5 (1)	0
Rash pustular	5 (1)	0
Skin lesion	5 (1)	0

Stomatitis	5 (1)	0
Any event leading to discontinuations of any study agent	147 (35)†	58 (14)
Most common events leading to discontinuations of any study agent§		
Infusion-related reaction	19 (5)	0
Paronychia	14 (3)	0
Rash	11 (3)	0
Pneumonia	8 (2)	3 (1)
Pulmonary embolism	8 (2)	2 (<1)
Pneumonitis	7 (2)	7 (2)
Dermatitis acneiform	6 (1)	0
Hypoalbuminemia	6 (1)	0
Interstitial lung disease	5 (1)	4 (1)
Asthenia	5 (1)	1 (<1)
Peripheral edema	5 (1)	0

*The safety population included all the patients who had undergone randomization and received at least one dose of any trial treatment.

†Protocol recommended dose modifications of amivantamab for grade ≥ 2 related toxicities first; by agent: 328 (78%), 193 (46%), and 145 (34%) interrupted, reduced, and discontinued amivantamab, respectively, and 299 (71%), 176 (42%), and 85 (20%) interrupted, reduced, and discontinued lazertinib, respectively.

‡Listed are adverse events that were reported in at least 3% of patients in any group.

§Listed are adverse events that were reported in at least 1% of patients in any group.

Table S11. Treatment-related Adverse Events*

	Amivantamab-lazertinib		Osimertinib	
Event — no. (%)	(n=421)		(n=428)	
Any event	414 (98)		378 (88)	
Grade ≥3	252 (60)		59 (14)	
Any serious event	97 (23)		24 (6)	
Any event resulting in death	4 (1)		2 (<1)	
Any event leading to:				
Interruptions	Amivantamab†: 282 (67) Lazertinib: 241 (57)		81 (19)	
Reductions	Amivantamab†: 184 (44) Lazertinib: 165 (39)		16 (4)	
Discontinuations	Amivantamab†: 100 (24) Lazertinib: 38 (9)		14 (3)	
Adverse events reported in ≥15% of patients in either group‡	All	Grade ≥3	All	Grade ≥3
Paronychia	285 (68)	44 (10)	115 (27)	2 (<1)
Infusion-related reaction	265 (63)	27 (6)	0	0
Rash	256 (61)	65 (15)	121 (28)	3 (1)
Hypoalbuminemia	164 (39)	17 (4)	11 (3)	0
Increased alanine aminotransferase	122 (29)	17 (4)	38 (9)	4 (1)
Dermatitis acneiform	122 (29)	34 (8)	54 (13)	0
Stomatitis	116 (28)	4 (1)	77 (18)	1 (<1)

Peripheral edema	112 (27)	6 (1)	10 (2)	0
Diarrhea	99 (24)	8 (2)	149 (35)	2 (<1)
Increased aspartate aminotransferase	95 (23)	9 (2)	34 (8)	1 (<1)
Pruritus	93 (22)	2 (<1)	63 (15)	1 (<1)
Decreased appetite	73 (17)	2 (<1)	40 (9)	1 (<1)
Nausea	68 (16)	2 (<1)	32 (7)	1 (<1)
Dry skin	65 (15)	1 (<1)	54 (13)	1 (<1)
Thrombocytopenia	57 (14)	1 (<1)	75 (18)	3 (1)

*The safety population included all the patients who had undergone randomization and received at least one dose of any trial treatment.

†Protocol recommended dose modifications of amivantamab for grade ≥ 2 related toxicities first.

‡Events in this category are listed according to decreasing incidence in the amivantamab-lazertinib group.

Table S12. All Grade 5 Adverse Events*

	Amivantamab-lazertinib	Osimertinib
Event — no. (%)	(n=421)	(n=428)
Pneumonia	5 (1)	4 (1)
Respiratory failure	4 (1)	2 (<1)
Sudden death	4 (1)†	1 (<1)
Death	3 (1)	2 (<1)
Myocardial infarction	3 (1)‡	0
Pulmonary embolism	2 (<1)	2 (<1)
Septic shock	2 (<1)	1 (<1)
COVID-19	1 (<1)	3 (1)
Pericardial effusion	1 (<1)	1 (<1)
Acinetobacter sepsis	1 (<1)	0
Acute respiratory distress syndrome	1 (<1)	0
Arteriosclerosis coronary artery	1 (<1)	0
Cardiopulmonary failure	1 (<1)	0
Cerebral infarction	1 (<1)	0
Circulatory collapse	1 (<1)	0
Coronary artery disease	1 (<1)§	0
COVID-19 pneumonia	1 (<1)	0
Ischemic cerebral infarction	1 (<1)	0

Myocardial rupture	1 (<1)	0
Pneumonitis	1 (<1)§	0
Urosepsis	1 (<1)	0
Dyspnea	0	3 (1)
Interstitial lung disease	0	2 (<1)§
Cardiac failure	0	1 (<1)
Cerebral hemorrhage	0	1 (<1)
Cerebrovascular accident	0	1 (<1)
Hemoptysis	0	1 (<1)
Ketoacidosis	0	1 (<1)
Lower respiratory tract infection	0	1 (<1)
Metabolic acidosis	0	1 (<1)
Pleural effusion	0	1 (<1)
Pneumonia aspiration	0	1 (<1)
Respiratory tract infection	0	1 (<1)
Sepsis	0	1 (<1)

*The safety population included all the patients who had undergone randomization and received at least one dose of any trial treatment. All grade 5 adverse events correspond to treatment-emergent adverse events leading to death and vice versa.

†One event in the amivantamab-lazertinib group was deemed related to any study treatment by the investigator.

‡Two events in the amivantamab-lazertinib group were deemed related to any study treatment by the investigator.

§Deemed as related to any study treatment by the investigator.

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